Shivling Bhange

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Profile

Frontend Developer with 5+ years of experience building scalable, high-performance web applications for automotive and enterprise security domains. Proven track record in reducing report generation time by 60-65% for large datasets and improving application load times by 2x through performance optimization and efficient data parsing. Skilled in React.js, Node.js, and TypeScript, known for delivering robust, maintainable software through TDD/BDD practices and clean component architecture.

Skills

Frontend: React.js, Redux, React Hooks, Context API, Angular, JavaScript (ES6+), TypeScript, HTML5, CSS3, MUI, Bootstrap, React Router, Axios, Webpack

Database: MongoDB

DevOps & Tools: Git, GitHub, Bitbucket, Jira, AWS (EC2, S3, Lambda), Docker Basics, Postman

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Testing: Jest, React Testing Library, Mocha, Chai, BDD (Gherkin), TDD, Unit Testing, Integration Testing

Other: CI/ CD basics, Agile/ Scrum, Unit Testing (Jest, Mocha)

Professional Experience

Software Engineer – – – BMW Techworks India (**BMW Group**) *01/2024 – Present | Pune, India*

- Improved system performance by reducing report generation time by 60-65% for 150-200MB datasets through memoization and pagination techniques, enabling faster data processing for engineering teams.
- Developed a high-precision ECU comparison tool using data structure & algorithms to optimize rendering pipelines, applying analytical precision to debugging complex parameter changes for 200+ engineers, reducing analysis time for 200K+ parameters to under 5 seconds.
- Parsed binary and calibration data into readable structures using Rust WASM and custom parsers, improving decoding accuracy and enabling real-time analysis of ECU parameters.
- Implemented Role-Based Access Control (RBAC) with Admin and Application Developer roles for 200+ users, ensuring secure access and user-level permissions in compliance with ASPICE and ISO standards.
- Built advanced UI tools including search, filters, and visual signal comparisons using React.js and WASM, improving workflow efficiency by 70% for 200+ engineers.
- Automated report generation, reducing manual review effort for ECU calibration validation improved accuracy and speed.
- Built a reusable React component library with 500+ unit and integration tests using Jest, React Testing Library, and Mocha following TDD/BDD practices, achieving 90% code coverage and significantly reducing regression bugs across the application.
- Designed and implemented a diff report to detect added, removed, and modified ECU parameters by comparing two DCM files, reducing manual analysis of 200K+ parameters from 2-3 hours to under 5 seconds.

Software Engineer – – Colortokens Inc *06/2022 – 01/2024 | Banglore, IND*

- Designed a centralized admin dashboard to manage and monitor 100K+ endpoints across the network, displaying hostname, IP, groups, firewall rules, policies, and system status, enabling real-time visibility and reducing manual monitoring effort.

- Integrated with X-Access VPN desktop application to enable secure token-based authentication, access control, and endpoint-level policy enforcement, streamlining onboarding for the entire user base while strengthening security through authenticator-based verification.
- Enabled administrators to view and manage firewall rules for all 100K+ onboarded systems from a centralized interface, eliminating the need to check each system individually and ensuring consistent network traffic control and unauthorized access blocking.
- Optimized REST API performance through indexing and query optimization using NodeJS, reducing initial load time for 200K+ records from 6 seconds to 3 seconds, achieving 2x faster data retrieval.
- Implemented remote resource decommissioning to securely offboard endpoints, automatically deleting system policies, access tokens, and associated data in the background, ensuring proper offboarding and compliance.
- Developed audit logs, alerts, role-based permissions, and alert history for security compliance, reducing audit log load time by 70% through API optimization.
- Provided real-time visibility into endpoint health, connection status, OS-level details, and threat alerts.

Software Engineer – GravityTech Software Solution *01/2021 – 06/2022* | *Pune, IND*

- Engineered secure and scalable internal web applications for a banking institution, ensuring compliance with industry-grade security protocols and data protection standards.
- Developed global access portals serving internal clients across multiple regions, enabling seamless cross-location collaboration and centralized access to critical banking resources.
- Delivered high-performance applications while ensuring strict adherence to internal security protocols and banking industry compliance standards.
- Implemented Multi-Factor Authentication (MFA) to enhance security during user registration and account creation in web and mobile applications, reducing unauthorized access risks.
- Created intuitive, data-driven UI dashboards that improved internal operations and enabled faster, more informed decision-making for banking teams.
- Designed and implemented front-end solutions that streamlined workflows and enhanced user experience, driving measurable improvements in operational efficiency.

Education

SGGS Institute of Engineering & Technology (Gov. Autonomous) *06/2019*

B.Tech. (Electronics & Telecommunication)

Certificates

Test-Driven Development for JavaScript

Introduction to Agile Development and Scrum